

Signals

- A **signal** is a small message that notifies a process that an event of some type has occurred in the system
 - akin to exceptions and interrupts
 - sent from the kernel (sometimes at the request of another process) to a process
 - signal type is identified by small integer ID's (1-30)
 - only information in a signal is its ID and the fact that it arrived

<i>ID</i>	<i>Name</i>	<i>Default Action</i>	<i>Corresponding Event</i>
2	SIGINT	Terminate	Interrupt (e.g., ctl-c from keyboard)
9	SIGKILL	Terminate	Kill program (cannot override or ignore)
11	SIGSEGV	Terminate & Dump	Segmentation violation
14	SIGALRM	Terminate	Timer signal
17	SIGCHLD	Ignore	Child stopped or terminated

Sending a Signal

- Kernel *sends* (delivers) a signal to a *destination process* by updating some state in the context of the destination process
- Kernel sends a signal for one of the following reasons:
 - Kernel has detected a system event such as divide-by-zero (SIGFPE) or the termination of a child process (SIGCHLD)
 - Another process has invoked the **kill** system call to explicitly request the kernel to send a signal to the destination process

Receiving a Signal

- A destination process *receives* a signal when it is forced by the kernel to react in some way to the delivery of the signal
- Three possible ways to react:
 - *Ignore* the signal (do nothing)
 - *Terminate* the process (with optional core dump)
 - *Catch* the signal by executing a user-level function called *signal handler*
 - Akin to a hardware exception handler being called in response to an asynchronous interrupt

Pending and Blocked Signals

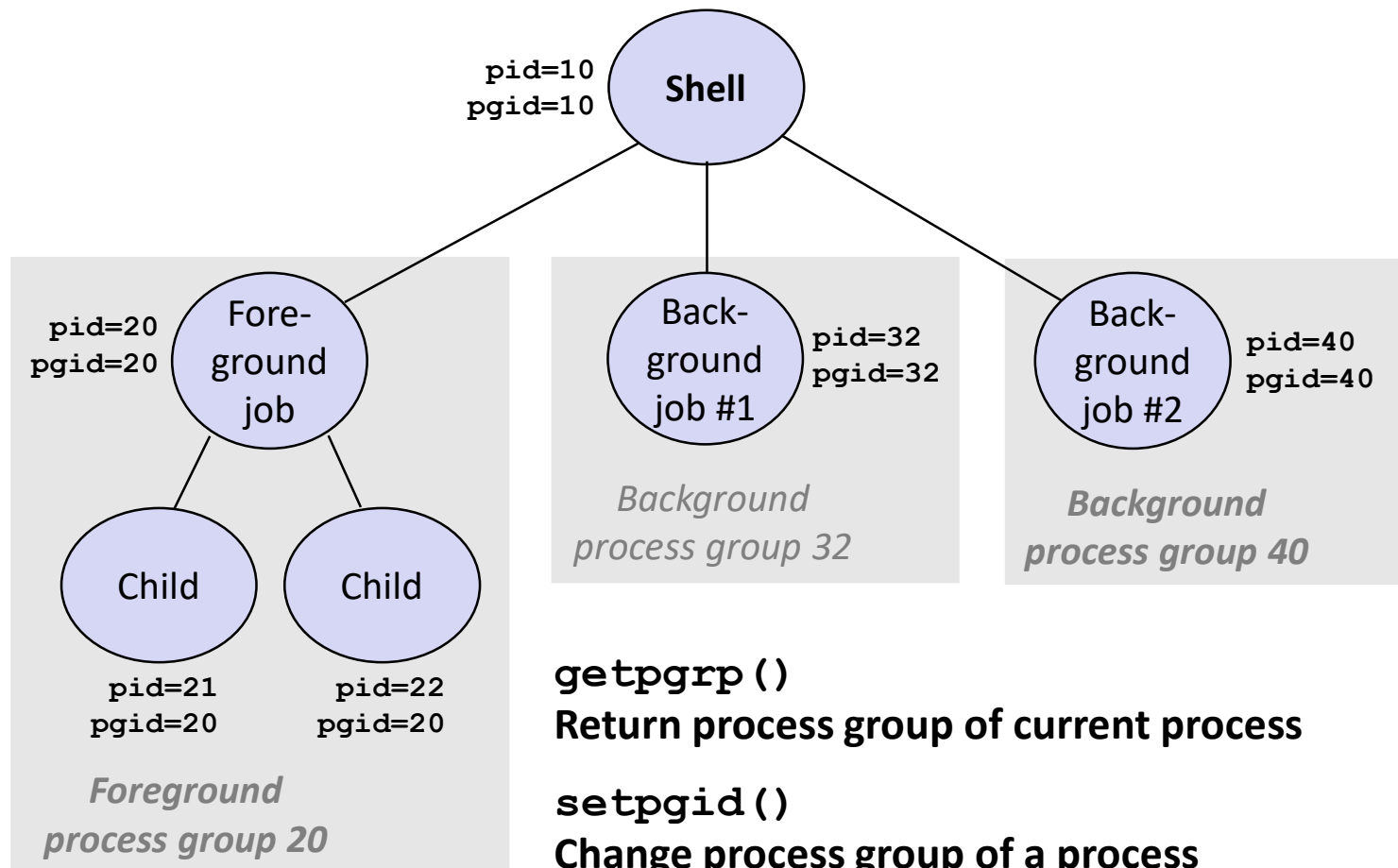
- A signal is *pending* if sent but not yet received
 - There can be at most one pending signal of any particular type
 - Important: Signals are not queued
 - If a process has a pending signal of type k, then subsequent signals of type k that are sent to that process are discarded
- A process can *block* the receipt of certain signals
 - Blocked signals can be delivered, but will not be received until the signal is unblocked
- A pending signal is received at most once

Signal Concepts

- Kernel maintains **pending** and **blocked** bit vectors in the context of each process
 - **pending**: represents the set of pending signals
 - Kernel sets bit k in **pending** when a signal of type k is delivered
 - Kernel clears bit k in **pending** when a signal of type k is received
 - **blocked**: represents the set of blocked signals
 - Can be set and cleared by using the **sigprocmask** function

Process Groups

- Every process belongs to exactly one process group



Sending Signals with `/bin/kill` Program

- `/bin/kill` program
sends arbitrary signal to a
process or process group

- Examples

- `/bin/kill -9 24818`
Send SIGKILL to process 24818
- `/bin/kill -9 -24817`
Send SIGKILL to every process
in process group 24817

```
linux> ./forks 16
Child1: pid=24818 pgrp=24817
Child2: pid=24819 pgrp=24817

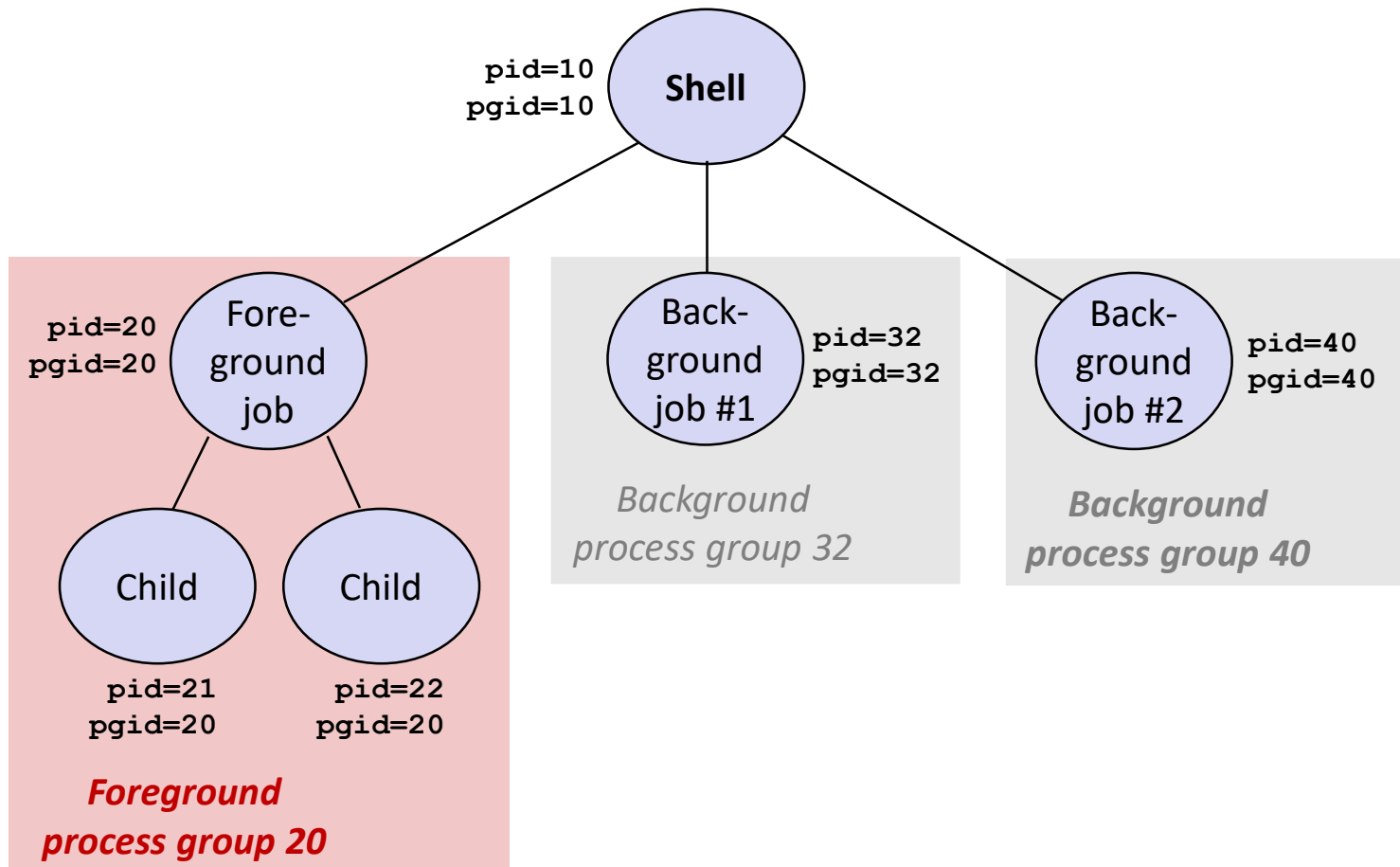
linux> ps
  PID TTY          TIME CMD
 24788 pts/2        00:00:00 tcsh
 24818 pts/2        00:00:02 forks
 24819 pts/2        00:00:02 forks
 24820 pts/2        00:00:00 ps
linux> /bin/kill -9 -24817
linux> ps
  PID TTY          TIME CMD
 24788 pts/2        00:00:00 tcsh
 24823 pts/2        00:00:00 ps
linux>
```

Analogue in Windows Taskkill

- **Standard (Requesting Exit):** `taskkill /pid ***` (functionally analog to SIGTERM).
- If the process has a GUI, it sends a WM_CLOSE message to the main window. The application message loop receives this and decides how to handle it (e.g., "Save changes?" prompt). Console applications must register a Control Handler.
- **Forceful:** `taskkill /f /pid ***` (functionally analog to SIGKILL).
- This bypasses all request mechanisms and calls the underlying kernel API `TerminateProcess()`. The kernel stops the threads and releases resources immediately.

Sending Signals from the Keyboard

- Typing ctrl-c (ctrl-z) sends a SIGINT (SIGTSTP) to every job in the foreground process group.
 - SIGINT – default action is to terminate each process
 - SIGTSTP – default action is to stop (suspend) each process



Example of `ctrl-c` and `ctrl-z`

```
bluefish> ./forks 17
Child: pid=28108 pgrp=28107
Parent: pid=28107 pgrp=28107
<types ctrl-z>
Suspended
bluefish> ps w
  PID TTY          STAT       TIME COMMAND
 27699 pts/8        Ss          0:00 -tcsh
 28107 pts/8        T           0:01 ./forks 17
 28108 pts/8        T           0:01 ./forks 17
 28109 pts/8        R+          0:00 ps w
bluefish> fg
./forks 17
<types ctrl-c>
bluefish> ps w
  PID TTY          STAT       TIME COMMAND
 27699 pts/8        Ss          0:00 -tcsh
 28110 pts/8        R+          0:00 ps w
```

STAT (process state) Legend:

First letter:

S: sleeping

T: stopped

R: running

Second letter:

s: session leader

+: foreground proc group

See “man ps” for more details

Sending Signals with kill Function

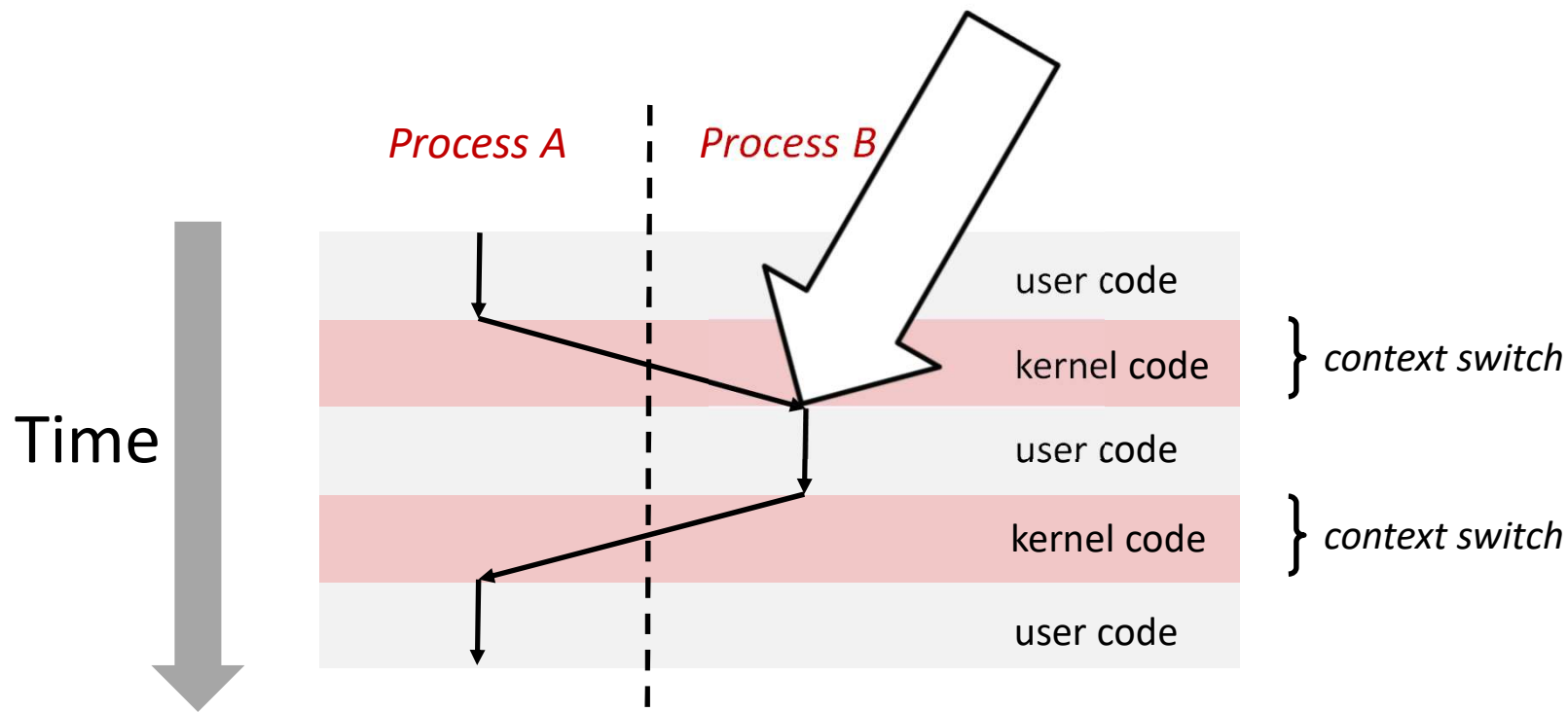
```
void fork12()
{
    pid_t pid[N];
    int i, child_status;
    for (i = 0; i < N; i++)
        if ((pid[i] = fork()) == 0)
            while(1); /* Child infinite loop */

    /* Parent terminates the child processes */
    for (i = 0; i < N; i++) {
        printf("Killing process %d\n", pid[i]);
        kill(pid[i], SIGINT);
    }

    /* Parent reaps terminated children */
    for (i = 0; i < N; i++) {
        pid_t wpid = wait(&child_status);
        if (WIFEXITED(child_status))
            printf("Child %d terminated with exit status %d\n",
                wpid, WEXITSTATUS(child_status));
        else
            printf("Child %d terminated abnormally\n", wpid);
    }
}
```

Receiving Signals

- Suppose kernel is returning from an exception handler and is ready to pass control to process p



Important: All context switches are initiated by calling some exceptional handler.

Receiving Signals

- Suppose kernel is returning from an exception handler and is ready to pass control to process p
- Kernel computes $\text{pnb} = \text{pending} \ \& \ \sim\text{blocked}$
 - The set of pending nonblocked signals for process p
- If $(\text{pnb} == 0)$
 - Pass control to next instruction in the logical flow for p
- Else
 - Choose least nonzero bit k in pnb and force process p to *receive* signal k
 - The receipt of the signal triggers some *action* by p
 - Repeat for all nonzero k in pnb
 - Pass control to next instruction in logical flow for p

Default Actions

- Each signal type has a predefined *default action*, which is one of:
 - The process terminates
 - The process terminates and dumps core
 - The process stops until restarted by a SIGCONT signal
 - The process ignores the signal

Installing Signal Handlers

- The `signal` function modifies the default action associated with the receipt of signal `signum`:
 - `handler_t *signal(int signum, handler_t *handler)`
- Different values for `handler`:
 - `SIG_IGN`: ignore signals of type `signum`
 - `SIG_DFL`: revert to the default action on receipt of signals of type `signum`
 - Otherwise, `handler` is the address of a *signal handler*
 - Called when process receives signal of type `signum`
 - Referred to as *“installing”* the handler
 - Executing handler is called *“catching”* or *“handling”* the signal
 - When the handler executes its return statement, control passes back to instruction in the control flow of the process that was interrupted by receipt of the signal

Signal Handling Example

```
void int_handler(int sig) {
    safe_printf("Process %d received signal %d\n", getpid(), sig);
    exit(0);
}
```

```
void fork13() {
    pid_t pid[N];
    int i, child_status;
    signal(SIGINT, int_handler);
    for (i = 0; i < N; i++)
        if ((pid[i] = fork()) == 0) {
            while(1); /* child infinite loop */
        }
    for (i = 0; i < N; i++) {
        printf("Killing process %d\n", pid[i]);
        kill(pid[i], SIGINT);
    }
    for (i = 0; i < N; i++) {
        pid_t wpid = wait(&child_status);
        if (WIFEXITED(child_status))
            printf("Child %d terminated with exit status %d\n",
                   wpid, WEXITSTATUS(child_status));
        else
            printf("Child %d terminated abnormally\n", wpid);
    }
}
```

```
linux> ./forks 13
```

```
Killing process 25417
```

```
Killing process 25418
```

```
Killing process 25419
```

```
Killing process 25420
```

```
Killing process 25421
```

```
Process 25417 received signal 2
```

```
Process 25418 received signal 2
```

```
Process 25420 received signal 2
```

```
Process 25421 received signal 2
```

```
Process 25419 received signal 2
```

```
Child 25417 terminated with exit status 0
```

```
Child 25418 terminated with exit status 0
```

```
Child 25420 terminated with exit status 0
```

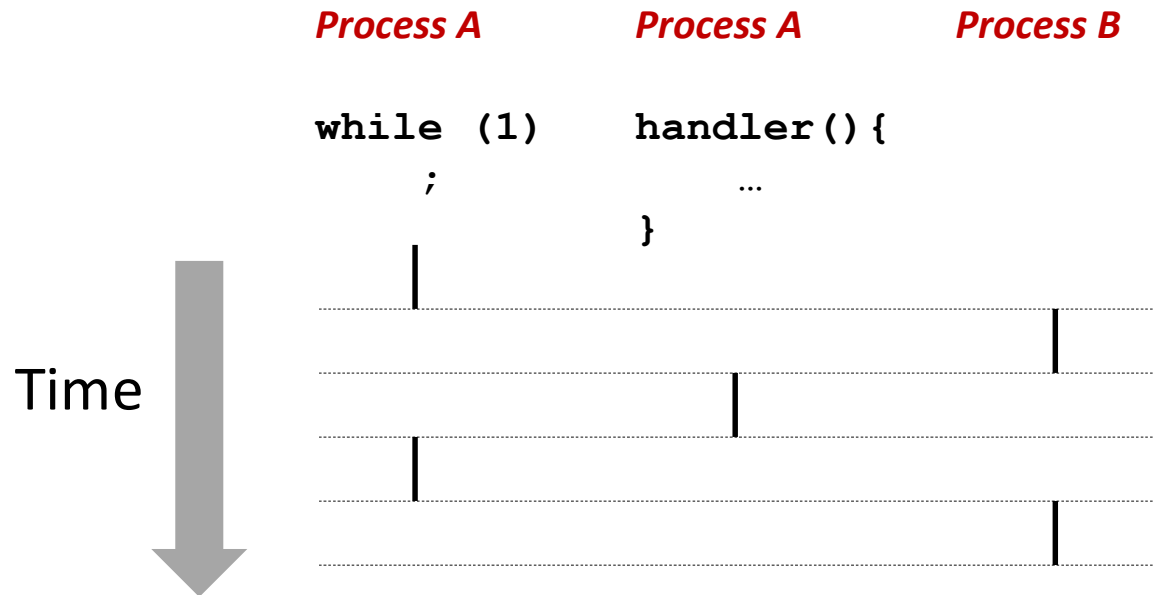
```
Child 25419 terminated with exit status 0
```

```
Child 25421 terminated with exit status 0
```

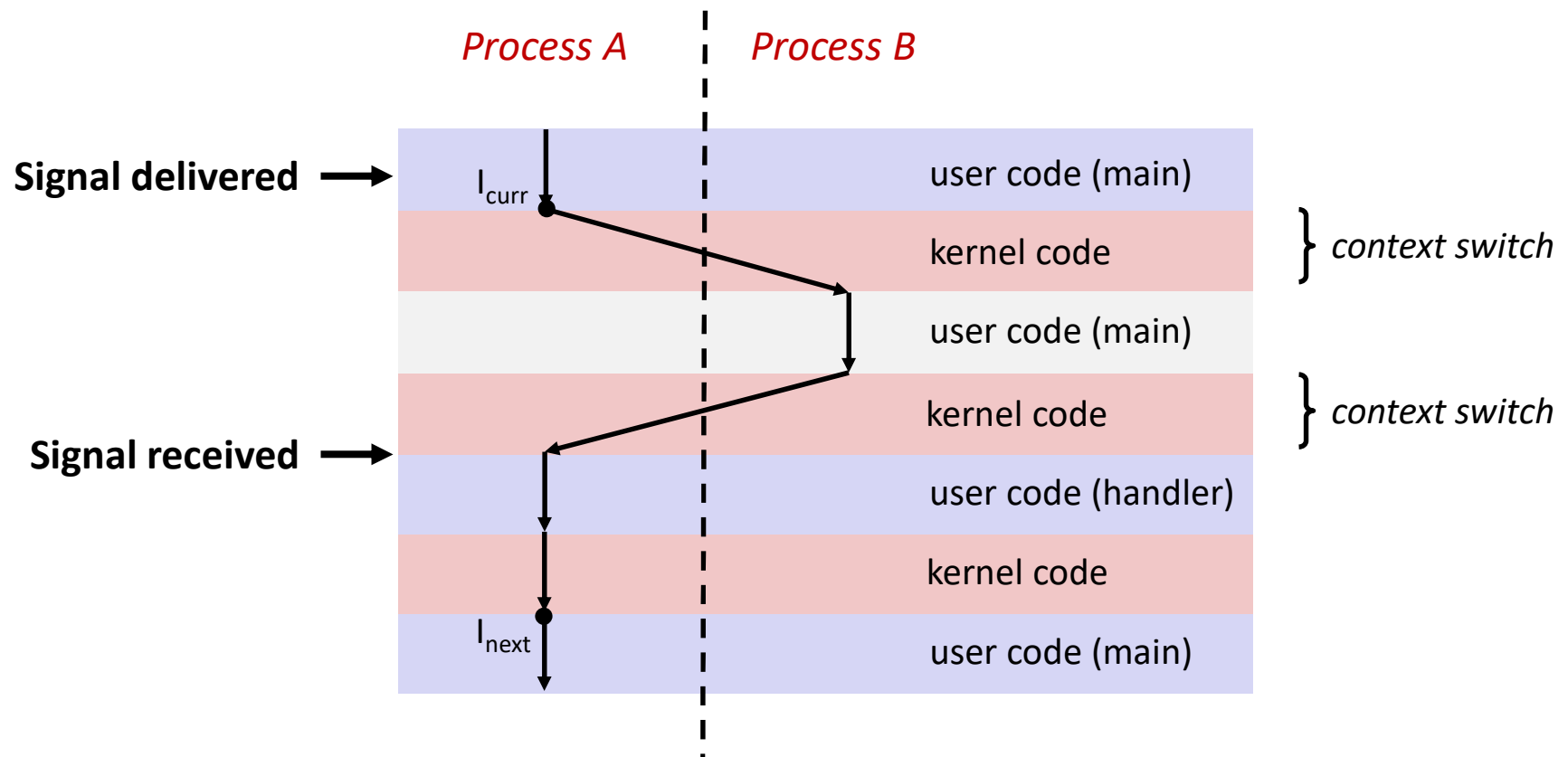
```
linux>
```


Signals Handlers as Concurrent Flows

- A signal handler is a separate logical flow (not process) that runs concurrently with the main program
 - “concurrently” in the “not sequential” sense



Another View of Signal Handlers as Concurrent Flows



Signal Handler Funkiness

```
int ccount = 0;
void child_handler(int sig)
{
    int child_status;
    pid_t pid = wait(&child_status);
    ccount--;
    safe_printf(
        "Received signal %d from process %d\n",
        sig, pid);
}

void fork14()
{
    pid_t pid[N];
    int i, child_status;
    ccount = N;
    signal(SIGCHLD, child_handler);
    for (i = 0; i < N; i++)
        if ((pid[i] = fork()) == 0) {
            sleep(1); /* deschedule child */
            exit(0); /* Child: Exit */
        }
    while (ccount > 0)
        pause(); /* Suspend until signal occurs */
}
```

■ Pending signals are not queued

- For each signal type, just have single bit indicating whether or not signal is pending
- Even if multiple processes have sent this signal

```
linux> ./forks 14
Received SIGCHLD signal 17 for process 21344
Received SIGCHLD signal 17 for process 21345
```

Living With Nonqueuing Signals

- Must check for all terminated jobs
 - Typically loop with `waitpid`

```
void child_handler2(int sig)
{
    int child_status;
    pid_t pid;
    while ((pid = waitpid(-1, &child_status, WNOHANG)) > 0) {
        ccount--;
        safe_printf("Received signal %d from process %d\n",
                    sig, pid);
    }
}
```

```
void fork15()
{
    . . .
    signal(SIGCHLD, child_handler2);
    . . .
}
```

```
greatwhite> forks 15
Received signal 17 from process 27476
Received signal 17 from process 27477
Received signal 17 from process 27478
Received signal 17 from process 27479
Received signal 17 from process 27480
greatwhite>
```

More Signal Handler Funkiness

- **Signal arrival during long system calls (say a `read`)**
- **Signal handler interrupts `read` call**
 - Linux: upon return from signal handler, the `read` call is restarted automatically
 - Some other flavors of Unix can cause the `read` call to fail with an `EINTR` error number (`errno`)
in this case, the application program can restart the slow system call
- **Subtle differences like these complicate the writing of portable code that uses signals**
 - Consult your textbook for details

A Program That Reacts to Externally Generated Events (Ctrl-c)

```
#include <stdlib.h>
#include <stdio.h>
#include <signal.h>

void handler(int sig) {
    safe_printf("You think hitting ctrl-c will stop the bomb?\n");
    sleep(2);
    safe_printf("Well...");
    sleep(1);
    printf("OK\n");
    exit(0);
}

main() {
    signal(SIGINT, handler); /* installs ctrl-c handler */
    while(1) {
    }
}
```

```
linux> ./external
<ctrl-c>
You think hitting ctrl-c will stop
the bomb?
Well...OK
linux>
```

external.c

A Program That Reacts to Internally Generated Events

```
#include <stdio.h>
#include <signal.h>

int beeps = 0;

/* SIGALRM handler */
void handler(int sig) {
    safe_printf("BEEP\n");

    if (++beeps < 5)
        alarm(1);
    else {
        safe_printf("BOOM!\n");
        exit(0);
    }
}
```

internal.c

```
main() {
    signal(SIGALRM, handler);
    alarm(1); /* send SIGALRM in
               1 second */

    while (1) {
        /* handler returns here */
    }
}
```

```
linux> ./internal
BEEP
BEEP
BEEP
BEEP
BEEP
BOOM!
bass>
```

Async-Signal-Safety

- Function is *async-signal-safe* if either reentrant (all variables stored on stack frame, CS:APP2e 12.7.2) or non-interruptible by signals.
- Posix guarantees 117 functions to be async-signal-safe
 - `write` is on the list, `printf` is not
- One solution: async-signal-safe wrapper for `printf`:

```
void safe_printf(const char *format, ...) {
    char buf[MAXS];
    va_list args;

    va_start(args, format);           /* reentrant */
    vsnprintf(buf, sizeof(buf), format, args); /* reentrant */
    va_end(args);                     /* reentrant */
    write(1, buf, strlen(buf));       /* async-signal-safe */
}
```


Structured Exception Handling (SEH) in Windows x64

- In 32-bit Windows, SEH was stack-based (fs:[0]), making it easy to exploit via buffer overflows. In x64 Windows, SEH is table-based (using .pdata and .xdata sections in the PE file). It is optimized for speed during normal execution.
- **Blocks:**
- **__try:** Defines a guarded block of code.
- **__except(...):** Catches exceptions.
- **__finally:** Defines code that always executes, regardless of whether an exception occurred or return was called. (Critical for resource cleanup, e.g., releasing critical sections).

Nonlocal Jumps: `setjmp/longjmp`

- **Powerful (but dangerous) user-level mechanism for transferring control to an arbitrary location**
 - Controlled to way to break the procedure call / return discipline
 - Useful for error recovery and signal handling
- **`int setjmp(jmp_buf j)`**
 - Must be called before `longjmp`
 - Identifies a return site for a subsequent `longjmp`
 - Called once, returns one or more times
- **Implementation:**
 - Remember where you are by storing the current *register context*, *stack pointer*, and *PC value* in `jmp_buf`
 - Return 0

setjmp/longjmp (cont)

- `void longjmp(jmp_buf j, int i)`
 - Meaning:
 - return from the `setjmp` remembered by jump buffer `j` again ...
 - ... this time returning `i` instead of 0
 - Called after `setjmp`
 - Called once, but never returns
- **longjmp Implementation:**
 - Restore register context (stack pointer, base pointer, PC value) from jump buffer `j`
 - Set `%eax` (the return value) to `i`
 - Jump to the location indicated by the PC stored in jump buf `j`

setjmp/longjmp Example

```
#include <setjmp.h>
jmp_buf buf;

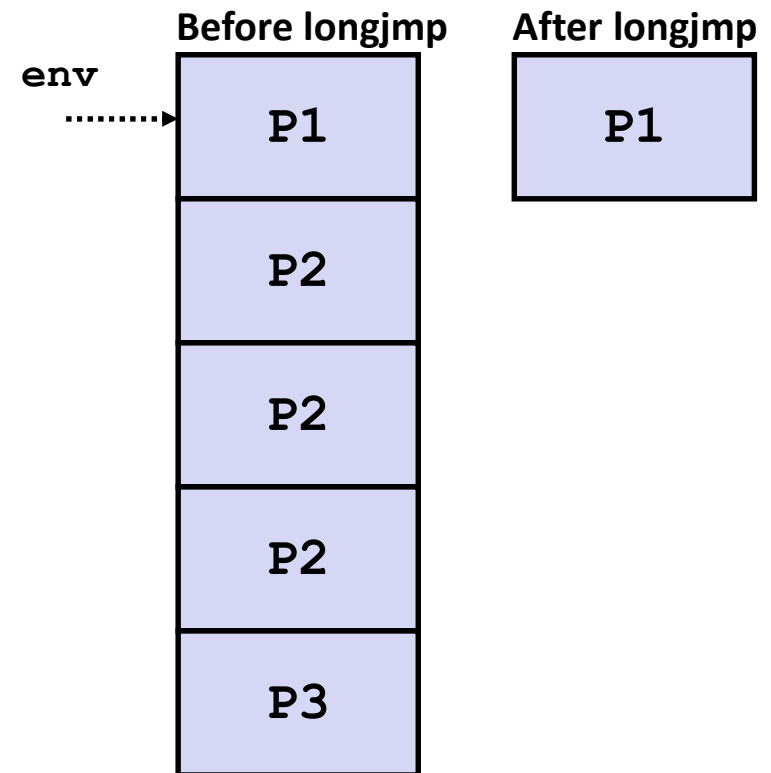
main() {
    if (setjmp(buf) != 0) {
        printf("back in main due to an error\n");
    } else {
        printf("first time through\n");
        p1(); /* p1 calls p2, which calls p3 */
    }
    ...
    p3() {
        <error checking code>
        if (error)
            longjmp(buf, 1)
    }
}
```

Limitations of Nonlocal Jumps

■ Works within stack discipline

- Can only long jump to environment of function that has been called but not yet completed

```
jmp_buf env;  
  
P1()  
{  
    if (setjmp(env)) {  
        /* Long Jump to here */  
    } else {  
        P2();  
    }  
}  
  
P2()  
{  
    . . . P2(); . . . P3();  
}  
  
P3()  
{  
    longjmp(env, 1);  
}
```

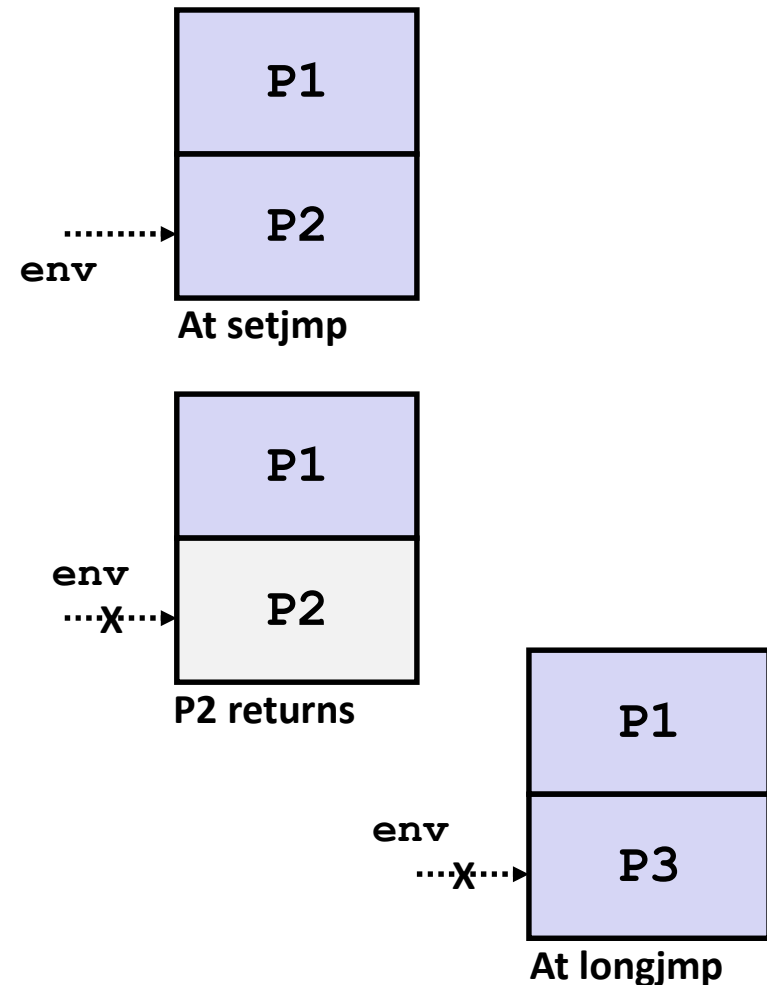


Limitations of Long Jumps (cont.)

■ Works within stack discipline

- Can only long jump to environment of function that has been called but not yet completed

```
jmp_buf env;  
  
P1()  
{  
    P2(); P3();  
}  
  
P2()  
{  
    if (setjmp(env)) {  
        /* Long Jump to here */  
    }  
}  
  
P3()  
{  
    longjmp(env, 1);  
}
```



Putting It All Together: A Program That Restarts Itself When `ctrl-c`'d

```
#include <stdio.h>
#include <signal.h>
#include <setjmp.h>

sigjmp_buf buf;

void handler(int sig) {
    siglongjmp(buf, 1);
}

main() {
    signal(SIGINT, handler);

    if (!sigsetjmp(buf, 1))
        printf("starting\n");
    else
        printf("restarting\n");

    while(1) {
        sleep(1);
        printf("processing...\n");
    }
}
```

```
greatwhite> ./restart
starting
processing...
processing...
processing...
restarting
processing... ← Ctrl-c
processing...
restarting
processing... ← Ctrl-c
processing...
processing...
```

restart.c

Summary

- **Signals provide process-level exception handling**
 - Can generate from user programs
 - Can define effect by declaring signal handler
- **Some caveats**
 - Very high overhead
 - >10,000 clock cycles
 - Only use for exceptional conditions
 - Don't have queues
 - Just one bit for each pending signal type
- **Nonlocal jumps provide exceptional control flow within process**
 - Within constraints of stack discipline

Multiprocessing and Signals

■ Low-Level OS Signals

```
import os
import signal
from multiprocessing import Process

def worker(pval):
    qval = pval + 3
p = Process(target=worker, args=(2,))
p.start()
# Sending a signal to the child
os.kill(p.pid, signal.SIGTERM)
```

To react to an OS signal, the child process must use the `signal` module to register a Handler.

`signal.signal(sig, handler)`: Associates a signal (like `SIGUSR1`) with a function that executes when the signal arrives.

Multiprocessing and Signals

- **multiprocessing.Event** (a thread-safe/process-safe boolean flag.)
 - `.set()` : Turns the flag to True (sends the "signal").
 - `.clear()` : Resets the flag to False.
 - `.wait()` : Blocks the receiving process until the flag becomes True.
 - `.is_set()` : Non-blocking check of the flag status.
- **multiprocessing.Condition** (Used for more complex signaling where a process needs to wait for a specific state change.)
 - `.notify()` / `.notify_all()` : Wakes up one or all processes waiting on this condition.
 - `.wait()` : Put the process to sleep until notified by another process.